Common Anemias

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CPhA strives to provide guidance for health-care professionals on the appropriate care of all individuals. However, we recognize that in this chapter, our content is based on clinical research studies of patients who may not reflect the identity of the individual patient presenting. Other resources exist that may be helpful, such as the World Professional Association for Transgender Health's (WPATH) Standard of Care. When the terms "women" and "men" are used in the text, they are used to maintain accuracy with what is reported in the clinical literature. We are aware these terms are not inclusive of all patients.

Introduction

Anemia is a condition that reduces the ability of the blood to carry oxygen to tissues. It is defined as having fewer than the normal number of red blood cells or less than the normal amount of hemoglobin (Hb) in the blood. Anemia is a significant global public health problem with an estimated 1.92 billion people affected.[1] The World Health Organization (WHO) recommends using Hb as a reliable objective indicator of diagnosing anemia, with thresholds set at an Hb level of <130 g/L in men and <120 g/L in nonpregnant women.[2] A reduction in hematocrit (Hct) or RBC count can also be used to define anemia. It is notable that these Hb ranges may not be applicable for all patients. Examples include people living at high altitudes, who have values higher than those living at sea level,[3] and people who smoke, who have higher Hct values than those who do not.[4] African-American patients may have Hb values 5–10 g/L lower than comparable Caucasian populations;[5] however, considering the high prevalence of anemia in African-American patients is primarily due to differences in social determinants of health,[6] the difference in Hb thresholds should be interpreted with caution.

In this review, anemias responsive to pharmacologic therapy are discussed. These include anemias in the adult population due to iron, vitamin B_{12} and/or folate deficiency, or anemias responding to erythropoietin therapy. As well, anemia of chronic disease is briefly discussed in the context of differential diagnosis from iron deficiency anemia.

Goals of Therapy

- Alleviate the signs and symptoms of anemia
- Restore normal or adequate Hb level
- Improve quality of life
- Prolong survival

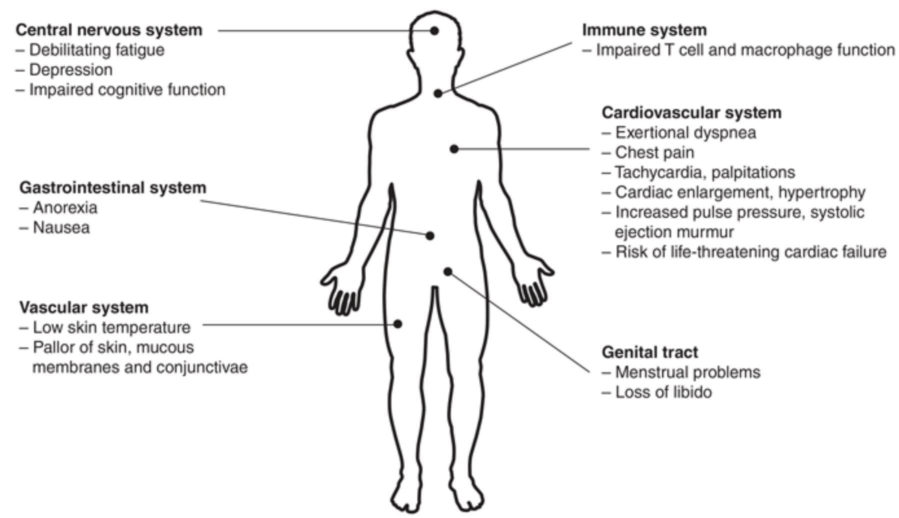
Investigations

- A history detailing presence of bleeding (including menstrual bleeding and blood donation), jaundice, GI symptoms, family history of anemia, medication use, diet, alcohol intake and comorbid conditions (chronic kidney disease, liver disease) is warranted.
- A physical examination may reveal the signs of anemia that occur when the oxygen-carrying capacity of the blood is unable to meet the oxygen requirements of body tissues (see Figure 1).[7] Additionally, there may be signs or symptoms that can point to the etiology of the anemia, e.g., glossitis and koilonychia in iron deficiency, paresthesia in vitamin B_{12} deficiency.
- Underlying cause(s) for anemia are variable and include:
  - deficiencies in dietary intake, e.g., iron, folate or vitamin B_{12} deficiency
  - conditions that are associated with inadequate absorption of vitamins by the GI system, e.g., gastric bypass, celiac disease, inflammatory bowel diseases
  - obvious or occult blood loss
  - hemolysis
  - inherited defects in hemoglobin (e.g., thalassemias, sideroblastic anemia) or the RBC membrane (e.g., glucose 6-phosphate dehydrogenase deficiency)
  - suppression of the bone marrow, e.g., due to medications including cytotoxic agents, antiretrovirals and folate antagonists, infectious causes, chronic alcohol use
  - primary diseases affecting the bone marrow, e.g., leukemia, myelodysplastic syndrome, aplastic anemia
  - infiltrative diseases affecting the bone marrow, e.g., cancer, sarcoidosis
  - deficiencies of erythropoietin from chronic kidney disease or comorbid conditions, e.g., hypothyroidism, hypogonadism
  - chronic inflammatory diseases (anemia of chronic disease)
- A CBC is required for the diagnosis of anemia. A diagnostic algorithm in Figure 2 is given as guidance and is based on the traditional classification of anemia according to red cell size, as reflected in the mean

corpuscular volume (MCV). The normal reference range for an MCV is 80–100 fL. A microcyte is a small RBC with MCV <80 fL, while a macrocyte is a large RBC with MCV >100 fL. A peripheral blood smear is also recommended during the initial evaluation of anemia, as the characteristics of RBCs (e.g., size, shape, size of the central pallor) and other blood cells can provide important clues for the diagnosis (e.g., sickling of RBCs in sickle cell anemia, megalocytes and segmented neutrophils in megaloblastic anemia).[8][9]

- Women, particularly those who are pregnant or of reproductive age, and children have a high prevalence of iron deficiency anemia, and screening should be considered. Postmenopausal patients and adult men presenting with iron deficiency anemia with no overt bleeding should be investigated for GI sources of blood loss.
- Ferritin is an iron-storage protein, and serum ferritin is an important test when screening for iron deficiency anemia. A ferritin level <30 mcg/L defines iron deficiency in adults and <20 mcg/L in children. Ferritin can be influenced by the presence of inflammation, infection, pregnancy or obesity and must be interpreted in the correct context.[10] In a state of chronic inflammation, a ferritin level <100 mcg/L defines iron deficiency[11] (note: thresholds for serum values vary by laboratory).
- Transferrin, an iron-transport protein, is also measured to determine iron status. An elevated level can indicate iron stores are low. The amount of iron that transferrin is capable of binding is called the total iron binding capacity (TIBC). A transferrin saturation percentage, the ratio of serum iron to total-iron binding capacity, provides useful information in determining iron metabolism. A reduced level (<20%) reflects iron deficiency while an elevated level (>50%) indicates iron overload.
- The serum folate level is prone to short-term fluctuations and may be misleading. The RBC folate level reflects time-averaged folate availability and is a more reliable indicator of tissue folate adequacy.[12]
- There is significant intra- and interindividual variation in serum cobalamin (vitamin B_{12}) levels, so patients with macrocytosis and borderline cobalamin levels (as defined by the local laboratory) need further assessment.[13] Cobalamin levels <200 pmol/L would generally be considered deficient or borderline deficient. Cobalamin levels can decrease during pregnancy without other evidence of deficiency.[14]
- Reticulocytes are immature red blood cells produced in the bone marrow. A reticulocyte count may be used in the assessment of normocytic anemia as a measure of bone marrow responsiveness to anemia. An elevated reticulocyte count (either as an absolute count or percentage) is indicative of acute hemolysis or bleeding, whereas an inappropriately low or normal reticulocyte count could suggest bone marrow suppression, infiltration or other abnormality.[15]

Figure 1: *Signs and Symptoms of Anemia*



Signs and symptoms of anemia occur when the oxygen-carrying capacity of the blood is unable to meet the oxygen requirements of body tissues. Adapted with permission of Elsevier Inc. from Ludwig H, Fritz E. Anemia in cancer patients. *Semin Oncol* 1998;25(3 Suppl 7):2-6. Copyright 1998; permission conveyed through Copyright Clearance Centre, Inc.

Iron Deficiency Anemia

Iron deficiency anemia (IDA) is the most common nutritional disorder, accounts for nearly half of all anemia cases, and is most prevalent among preschool children and women. Iron deficiency anemia is present in 1–2% of adults; iron deficiency without anemia occurs in up to 11% of women and 4% of men.[16] The most common cause of iron deficiency is blood loss, which can be overt or occult.[17] Blood loss through the GI tract is the most common cause of occult blood loss in men and in postmenopausal patients. Regular blood donors, patients with chronic hemolysis and patients who are menstruating may develop iron deficiency. Decreased iron absorption in patients due to disorders such as celiac disease or autoimmune atrophic gastritis or in those who have undergone gastric bypass surgery can also result in iron deficiency and IDA.

IDA typically presents as a microcytic anemia with a low MCV. A serum ferritin level is used in the diagnosis of IDA, and decreased levels are diagnostic. Other measurements (transferrin saturation) can also be used. Iron metabolism is regulated by hepcidin, a hormone that is primarily produced in the liver.[18] High levels of hepcidin promote the storage of iron in the reticuloendothelial cells of the liver, spleen, and bone marrow, and reduces iron absorption from the gut. The production of hepcidin is suppressed in response to hypoxia, iron deficiency and in the presence of erythropoietin. An increased level of hepcidin correlates with an elevated serum ferritin, which indicates iron stores are high. A patient with IDA will present with low ferritin, microcytic RBC, low Hb, and theoretically low hepcidin levels.[19] Measurement of hepcidin levels is currently not widely available outside of research settings.

Anemia of Chronic Disease

Anemia of chronic disease (ACD) is the second-most-prevalent cause of anemia after iron deficiency anemia.[11] During periods of inflammation or infection, hepcidin levels are elevated, as its production is stimulated by inflammatory cytokines such as interleukin-6 and bacterial lipopolysaccharide.[20] Hepcidin has been identified as an antimicrobial and acts to sequester iron from invading bacteria by removing iron from the circulation.[18] An elevated ferritin level with evidence of inflammation (e.g., increased serum C-reactive protein levels) in a patient with a normocytic/microcytic anemia may be indicative of ACD. Serum transferrin and transferrin saturation are laboratory measurements used to assess iron disorders (see Table 1).

The diagnosis and treatment of ACD can be a challenge, as there may be overlap of an iron deficiency state with ACD. As ACD is caused by an underlying disorder, it may not be feasible to completely eradicate the condition. If possible, addressing and controlling the inflammatory disorder is the recommended strategy to resolve ACD.

Table 1: *Laboratory Measurements in the Evaluation of Iron Status*[11][18][19]

Clinical Parameter	Iron Deficiency Anemia	Anemia of Chronic Disease	Iron Deficiency Anemia and Anemia of Chronic Disease
C-Reactive Protein[a]	Normal	High	High
Hemoglobin	Low	Low	Low
Ferritin[a]	Low	Normal-high	Normal-high
Transferrin Saturation	Low	Low-normal	Low-normal
Transferrin	High	Low-normal	Low
Mean Corpuscular Volume	Low	Low-normal	Low

[a] Acute phase reactant—level fluctuates rapidly in response to inflammation and tissue injury. Interpret cautiously, as it does not discriminate between acute and chronic inflammation; increased levels are seen in infection, trauma, autoimmune and inflammatory diseases states, and cancer.

Therapeutic Choices

Nonpharmacologic Choices

Dietary iron, especially from foods rich in heme iron (liver; lean red meats; seafood such as oysters, clams, tuna, salmon, sardines and shrimp), can contribute to the treatment of iron deficiency anemia, but typically requires more time to replete iron stores compared to pharmacologic replacement therapy.[21] Furthermore, it may not be sufficient in the face of more severe deficiency or in cases where iron losses exceed dietary intake. Plant-based diets contain nonheme iron, which has low bioavailability. Sources of nonheme iron include beans, lentils, tofu, spinach and fortified cereals.

The role of ascorbic acid (**vitamin C**) in the absorption of nonheme iron is controversial and a randomized controlled study and subsequent systematic review and meta-analyses have not shown any benefit from vitamin C supplementation in patients with iron deficiency anemia receiving oral iron supplementation.[22][23][24] It is unclear whether these findings can be extrapolated to nonheme iron obtained from dietary sources. It may be reasonable to recommend the combination of vitamin C–rich foods with plant-based sources of iron, especially in individuals who follow a vegetarian or vegan diet.[25]

Polyphenols and phytates found in tea and coffee can inhibit nonheme iron absorption.

Pharmacologic Choices

Additional information on drug therapy for iron deficiency anemia can be found in Table 2.

- Simple oral **iron salts** are the mainstay of iron supplementation therapy in most circumstances of iron deficiency anemia.
  - a variety of salts are available, with differing amounts of elemental iron per tablet
  - when iron salts are given on an empty stomach, absorption is enhanced, but side effects (including nausea and epigastric pain) are more common
  - recommended doses for iron supplementation vary based on the patient population and whether iron is used for prevention or treatment of iron deficiency
  - iron supplementation taken on alternate days as a single dose, compared to daily divided doses, may increase iron absorption and is associated with fewer gastrointestinal side effects[26]
    - a small trial in menstruating nonanemic, iron-deficient patients found alternate-day dosing of low iron doses (40–80 mg elemental iron) or intermittent dosing maximized the fraction of iron absorbed; iron supplementation given in the morning reduced the amount of further iron absorbed for up to 24 hours[27]

    - in a subsequent trial in women with iron deficiency anemia, high iron doses of 100 and 200 mg on alternate days resulted in higher fractional iron absorption[28]
    - a larger randomized trial found that alternate-day dosing was not associated with a higher increase in ferritin, but was associated with reduced iron deficiency (defined as ferritin <15 mcg/L or soluble transferrin receptor [sTfR] >8.3 mcg/mL) at 6 months and fewer GI side effects[29]
  - need for ongoing treatment after iron stores are replenished depends on the cause for the iron deficiency and whether there are ongoing losses
- **Polysaccharide-iron complex** contains 150 mg elemental iron/capsule. It may be better tolerated than oral iron salts, but reports are inconsistent. In a randomized trial of iron deficient children and infants, low-dose iron salts resulted in a greater increase in hemoglobin than polysaccharide-iron complex.[30]
- **Heme iron polypeptides** contains 11 mg elemental iron/tablet. It may be better absorbed and tolerated than oral iron salts. The recommended dosage is 1 tablet 3 times daily as a preventative dose for iron deficiency anemia. Evidence regarding its efficacy in the treatment of iron deficiency anemia is lacking.[31] Heme iron polypeptide is made from hydrolysis of bovine (animal) hemoglobin, which may be a consideration prior to its use.
- **Ferric maltol** contains 30 mg elemental iron/capsule. The recommended dosage is 1 capsule twice daily on an empty stomach. It is formulated to increase absorption in the small intestine.
- **Parenteral iron** is reserved for patients with malabsorption, who are intolerant to oral iron therapy (where ongoing losses exceed the capacity of the GI tract to absorb oral iron) or in clinical situations where large doses of iron are required in a short time period. The newer formulations have a low risk of serious adverse events (anaphylaxis) compared to the high molecular weight iron dextran preparations that are no longer on the market.[32]
  - parenteral iron may result in a more rapid rise in Hb compared to oral iron salts,[33] but requires administration in a hospital or outpatient clinic
  - parenteral iron consists of iron complexed with a carbohydrate or salt
  - high-dose parenteral iron preparations (ferric derisomaltose [known as iron isomaltoside in Europe] or ferric carboxymaltose) can provide a full iron replacement dose in 1–2 infusions
  - parenteral iron may reduce the need for RBC transfusion, but may also increase the risk of infection[34]

A reticulocyte response should be evident within 1 week of beginning iron therapy, with subsequent improvement in the Hb of about 10 g/L every 7–10 days.

There is no specific duration for iron supplementation. If an underlying cause of anemia is identified and treated, iron supplementation can be continued until iron stores are replenished. Some clinicians recommend continuing treatment for 3 months after the target Hb has been reached. If an underlying cause cannot be identified or cannot be definitively treated, it may be reasonable to educate the patient on the signs and symptoms of anemia and to routinely screen for recurrence of anemia through blood testing.

Choices during Pregnancy and Breastfeeding

It is recommended that pregnant patients meet the dietary requirements of iron through diet and/or supplements to prevent the development of iron deficiency during pregnancy and postpartum. During pregnancy, iron requirements increase due to expansion of maternal red cell mass and growth of the fetus and placenta.

Treatment of iron deficiency in pregnant patients is the same as in nonpregnant patients. There are no reported teratogenic effects of iron supplementation on the fetus. Iron normally passes into breast milk where it is an important source of dietary iron for the developing infant. The amount of iron in breast milk is generally not influenced significantly by maternal iron status.

In a trial of pregnant patients, 20 mg/day of elemental iron started at 20 weeks' gestation was well tolerated and resulted in fewer cases of iron deficiency and iron deficiency anemia.[35] Iron supplementation may be associated with lower rates of adverse pregnancy outcomes.[36][37]

- Oral **iron preparations** are the mainstay of iron supplementation therapy in most pregnant patients.
  - pregnant patients who experience nausea and GI symptoms during pregnancy can attempt to take lower doses of iron and/or take the iron with meals
  - another option for patients who do not tolerate daily iron is **intermittent iron supplementation**; limited data suggest intermittent iron supplementation (taken once, twice or three times weekly on nonconsecutive days) in pregnant patients is as effective in preventing iron deficiency and is better tolerated compared to daily regimens[38]
- **Parenteral iron** is reserved for pregnant patients with malabsorption or true intolerance to oral iron therapy, or where ongoing blood loss exceed the capacity of the GI tract to absorb oral iron resulting in severe anemia.
  - a systematic review and meta-analysis of 15 trials enrolling patients with postpartum anemia showed higher Hb concentrations at 6 weeks with parenteral compared to oral iron[39]
  - parenteral iron products have specific warnings and precautions for use in pregnancy; check the up-to-date product monograph before use

A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- The underlying cause of anemia should be investigated.
- The most common cause of microcytic anemia in North America is iron deficiency. In men and postmenopausal patients, occult GI bleeding should be ruled out, particularly since iron deficiency can be the first presentation of a GI malignancy (see Suggested Readings). Consideration should be given to current medications that may cause GI bleeding, e.g., NSAIDs, anticoagulants, antiplatelets.

- If there is no obvious GI blood loss, consider screening patients for celiac disease, which results in decreased absorption of iron from the GI tract.
- Pregnant patients can develop a decrease in Hb that is physiologic; however, iron deficiency is a common cause of anemia in this population and must be ruled out.
- If the Hb fails to respond as anticipated, consider that there may be:
  - ongoing blood loss
  - use of other medications that impair iron absorption (see Table 2)
  - a different or concurrent cause of anemia and/or an impaired erythropoietic response
  - adherence issues (GI side effects are the most common reason for nonadherence)
- Use a graduated approach to iron salt dosing and consider single doses on alternate days, initially taken with meals to limit GI upset.
- Small oral doses may be adequate in patients who are susceptible to GI upset. In older patients, daily doses of elemental iron as low as 15–50 mg are effective in the treatment of iron deficiency anemia.[32]
- Iron contained in enteric-coated tablets is poorly absorbed. These products should be avoided.

Megaloblastic Anemias

Megaloblastic anemias arise because of impaired DNA synthesis of the RBC caused by deficiencies of vitamin B_{12} (cobalamin) or folate (folic acid), or due to impaired DNA and RNA metabolism (drugs, myelodysplasia). Megaloblastic anemias are a subset of macrocytic anemias (anemia associated with increased red cell size) and are characterized by hypersegmented neutrophils on the peripheral blood film.

Patients with vitamin B_{12} deficiency may present with anemia/macrocytic RBCs or pancytopenia and may have neurologic complications including dementia, weakness, sensory neuropathy and paresthesias (subacute combined degeneration of the spinal cord). Folic acid supplementation may partially alleviate and mask the hematologic effects of cobalamin deficiency, but does not treat the neurologic complications. Patients with low-normal or even normal serum vitamin B_{12} values may be deficient and respond to vitamin B_{12} replacement.[40]

The clinical manifestations of megaloblastosis may be subtle in older adults and may precede the development of anemia. A high index of suspicion is warranted.[41]

Therapeutic Choices

Nonpharmacologic Choices

Restoring normal dietary intake of vitamin B_{12} and folate may be sufficient to completely reverse megaloblastosis. However, patients with neurologic deficits due to vitamin B_{12} deficiency should be treated pharmacologically to maximize the likelihood of full neurologic recovery. Vitamin B_{12} is present in foods of animal origin; examples include meat, fish, poultry, eggs and dairy products. Folate is present in a variety of foods including vegetables (especially dark green leafy vegetables), fruit, nuts and beans, and is also present in foods containing vitamin B_{12}: meat, fish, poultry, eggs and dairy products.

Abstinence from alcohol may be necessary.

Pharmacologic Choices

Additional information on drug therapy for megaloblastic anemia can be found in Table 2.

Vitamin B_{12}

- **Vitamin B_{12}** deficiency is treated with either **cyanocobalamin** (synthetic form of vitamin B_{12}) or **methylcobalamin** (naturally occurring form of vitamin B_{12}). While the dietary reference intake for vitamin B_{12} is 2.4 mcg/day, a daily dietary intake between 6–10 mcg ensures the maximal plasma vitamin B_{12} concentration in conditions of normal absorption.[42] Cobalamin replacement for treatment of pernicious anemia is in doses of 1000 mcg. Doses greater than 100 mcg/day exceed the physiologic binding capacity, but the excess is not toxic and is readily excreted by the kidneys. For this reason, there is a tendency to give more rather than less cobalamin, especially in patients with neurologic deficits.
- Vitamin B_{12} has traditionally been given parenterally because deficiency is frequently due to malabsorption, and most cases of malabsorption are attributable to pernicious anemia with its lack of intrinsic factor.
- High-dose oral vitamin B_{12} (e.g., 1000 mcg/day, followed by 1000 mcg/week and then 1000 mcg/month) therapy is effective,[43] feasible[44][45] and cost-effective,[46] but limitations include patient adherence and the need for more attentive monitoring.[47]
- A suggested approach is to:
  - administer parenteral vitamin B_{12} until all neurologic symptoms and hematologic abnormalities resolve
  - provide maintenance therapy by the route that best fits the patient’s circumstances; if oral vitamin B_{12} is used, follow-up with the patient to monitor Hb and ensure adherence
- Where dietary deficiency is clearly the cause of the vitamin B_{12} deficiency and the patient has no neurologic deficits, oral supplementation is adequate.
- It is important that patients with vitamin B_{12} deficiency are *not* treated with folic acid alone because this improves hematologic parameters but potentially worsens neurologic symptoms, which may become permanent.

Folic Acid

- Folate deficiency is treated with **folic acid**. The daily requirement for disease prevention is 200 mcg. Folate replacement is used in doses of 1–5 mg daily.
- Folic acid should be given for confirmed folate deficiency, pregnancy or in situations of increased demand, such as hemolysis. Certain drugs may cause folate deficiency, including methotrexate, phenytoin and sulfasalazine. A prescription of at least 5 mg folic acid per week is strongly recommended when methotrexate is taken regularly.[48]
- Prophylaxis with folic acid beginning 3 months before and during pregnancy is strongly recommended for the prevention of neural tube defects (see Nutritional Supplements for specific dosing recommendations).
- The oral route is sufficient to correct folate deficiency, even in patients with malabsorption syndromes.
- It is important to ensure that patients do not have concomitant vitamin B_{12} deficiency before folic acid replacement is started, since folate does not treat the neurologic manifestations of vitamin B_{12} deficiency.

Choices during Pregnancy and Breastfeeding

It is recommended that pregnant patients meet the dietary requirements of vitamin B_{12} and supplement with folic acid during pregnancy. With the exception of strict vegans, some vegetarians and patients with malabsorption (pernicious anemia), vitamin B_{12} is maintained at sufficient levels during pregnancy and does not typically require supplementation. Due to the recognized association between folate deficiency and fetal neural tube defects, folic acid supplementation is universally recommended before and during pregnancy. There are no teratogenic effects of either vitamin B_{12} or folic acid during pregnancy. Most authorities recommend that pregnant patients take a multivitamin containing folic acid during pregnancy and breastfeeding. Multivitamins recommended during pregnancy typically contain higher doses of folic acid than regular multivitamins. Both vitamin B_{12} and folate pass into breast milk but are compatible with breastfeeding.

- **Vitamin B_{12}** supplementation is reserved for pregnant patients with malabsorptive conditions (such as pernicious anemia) or for patients who are strict vegans. Patients who are vegetarians may become deficient during pregnancy due to increased demand. Supplementation is done in an identical manner to nonpregnant patients.
- **Folic acid** supplementation is universally recommended. The recommended supplemental dosage is based upon the pregnant individual and the sperm-producing partner’s risk for a neural tube defect or other folic acid–sensitive congenital anomaly (see Nutritional Supplements).

A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- In patients who are anemic due to vitamin B_{12} or folate deficiency, a reticulocyte response should be evident within 3–4 days of beginning vitamin B_{12} or folic acid therapy, with improvement in the Hb level by about day 10. Full resolution of the anemia should occur within about 2 months.
- The rapid production of new hematopoietic cells leads to a potentially dramatic shift of potassium from extracellular to intracellular compartments, which may cause profound hypokalemia.[49]
  - older patients on diuretic therapy for heart failure are at particular risk
  - in at-risk patients, obtain a baseline potassium level and give potassium supplementation in patients with low or borderline potassium levels; monitor potassium levels carefully in the first few days of therapy and adjust supplementation accordingly
- If the Hb fails to respond as anticipated, consider that there may be:
  - a different or concurrent cause of anemia and/or an impaired erythropoietic response
    - concomitant iron deficiency may show itself with an MCV that shifts from the macrocytic to the microcytic range
  - adherence issues, particularly in patients using oral supplements.
- Neurologic deficits may take 6 months or more to resolve; some severe deficits may persist.

Anemias Responsive to Pharmacologic Stimulation of Erythropoiesis

Most patients with nutritional anemias (iron, vitamin B_{12} or folate deficiency), hemolysis or bleeding have elevated levels of endogenous erythropoietin; however, there are a number of situations in which pharmacologic stimulation of red cell production using erythropoiesis-stimulating agents (ESAs) is beneficial, including the management of:

- Anemia secondary to chronic kidney disease
- Chemotherapy-induced anemia in patients with nonhematologic cancers[50][51]
- Anemia in the preoperative setting (surgical patients)[50]
- Symptomatic anemia in patients with low-risk myelodysplastic syndrome[52]
- Anemia due to antiretroviral therapy in patients with HIV infection[53]
- Anemia in patients with chronic hepatitis C receiving ribavirin[54]

Investigations

In most patients, Hb should be <100 g/L in order to qualify for ESA therapy. As well, selection of patients for ESA therapy is determined in part by baseline endogenous erythropoietin levels (approximately 3–30 units/L in healthy individuals). ESA therapy is generally titrated to ensure Hb levels do not exceed a certain threshold, depending on the indication.

It is important to ensure an adequate iron supply in conjunction with erythropoietin use.

Therapeutic Choices

Pharmacologic Choices

Additional information on erythropoiesis-stimulating agents can be found in Table 3.

The choice of agent and dosage regimen varies according to the clinical situation. Both available agents in Canada may be given by IV or SC injection. **Epoetin alfa** is a recombinant human erythropoietin with a relatively short half-life that is typically given at least 3 times per week. It may also be administered daily in the preoperative surgical setting where a more rapid rise is desired. **Darbepoetin alfa** is a synthetic erythropoietin analogue with a longer half-life that is typically given weekly or biweekly, and monthly in some patients.

Although ESAs have improved the quality of life in patients with anemia due to chronic renal failure,[55] they did not improve quality of life or fatigue in patients with anemia due to AIDS or cancer.[56][57] Additionally, the use of ESAs increases the risk of serious cardiovascular events when dosed to achieve a normal Hb.[58] Clinicians should carefully weigh the risks and benefits of ESAs before initiating their use. If ESAs are initiated, target Hb should be <100–120 g/L, depending on the indication, and Hb should be monitored at least monthly.[59]

ESAs have been associated with the development of pure red cell aplasia (PRCA), a potentially devastating complication in which neutralizing antibodies to the exogenous protein cross-react with endogenous erythropoietin, resulting in profound anemia. Changes in the formulation and handling of these proteins have greatly reduced the risk of PRCA.[60] Nevertheless, the possibility of PRCA should be considered if the patient’s anemia becomes refractory to therapy.

Choices during Pregnancy and Breastfeeding

There are no adequate and well-controlled studies using ESAs in pregnant patients. Use of ESAs in pregnancy does not seem to present a major risk to the fetus.[61] Worsening of hypertension or thrombosis are serious potential risks. However, anemia and frequent blood transfusions also present risks to the fetus. In these situations, ESAs should be given only if the potential benefits justify the potential risk to the fetus. Adequate iron supplementation is particularly important in this setting if ESAs are used.

ESAs have been used in infants with no serious adverse effects.[61] It is not known whether ESAs pass into breast milk; exercise caution in breastfeeding patients.

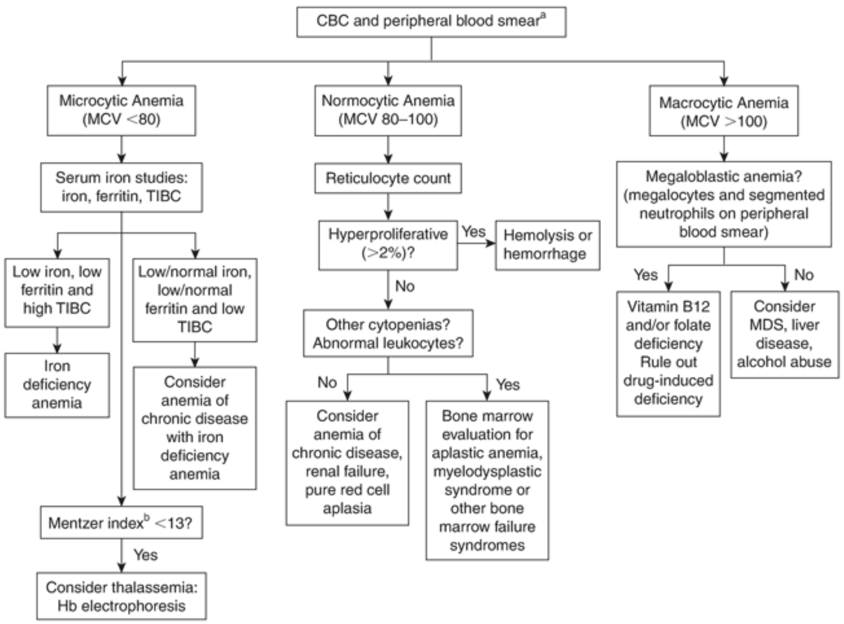
A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- ESAs have significant potential for toxicity and adverse consequences; they should be used judiciously in patients who would otherwise require transfusion support.
- Doses are titrated to achieve a gradual improvement in anemia without overshooting the target Hb.
- Rapid and/or excessive correction of anemia may provoke hypertension and seizures in susceptible individuals; erythrocytosis may predispose patients to thrombotic complications. Monitor blood pressure 3 times per week initially, and after each dose thereafter.

Algorithms

Figure 2: *Algorithm for the Assessment of Anemia*



[a] Peripheral blood smear serves to evaluate characteristics of red blood cells (e.g., size, shape, size of the central pallor) and other blood cells.
[b] Mentzer index = MCV/RBC

Abbreviations: CBC = complete blood count; Hb = hemoglobin; MCV = mean cell volume; MDS = myelodysplastic syndrome; RBC = red blood cell; TIBC = total iron-binding capacity

Drug Tables

Table 2: Drugs for the Treatment of Iron Deficiency and Megaloblastic Anemias

Drug/Cost[a]	Dosage	Adverse Effects	Comments
Drug Class: Iron Supplements, oral			
ferric maltol Accrufer $100	30 mg elemental iron BID PO (30 mg elemental iron/231.5 mg ferric maltol)	GI: flatulence, diarrhea, constipation, abdominal pain.	There is no evidence that one preparation is more effective than another. Avoid enteric-coated preparations. Vitamin C may enhance absorption of nonheme iron. Absorption is decreased by food and certain drugs: antacids, calcium carbonate, cholestyramine, levodopa, methyldopa, penicillamine, quinolones, sodium bicarbonate, tetracyclines. Separate administration by ~2 h.
ferrous fumarate Palafer, generics $5–10	105–200 mg elemental iron daily in 3 divided doses (100 mg elemental iron/300 mg ferrous fumarate)	GI: nausea, dyspepsia, constipation and/or diarrhea. Ameliorated by stepwise initiation of therapy (see text).	There is no evidence that one preparation is more effective than another. Avoid enteric-coated preparations. Vitamin C may enhance absorption of nonheme iron. Absorption is decreased by food and certain drugs: antacids, calcium carbonate, cholestyramine, levodopa, methyldopa, penicillamine, quinolones, sodium bicarbonate, tetracyclines. Separate administration by ~2 h.
ferrous gluconate generics < $5	105–200 mg elemental iron daily in 3 divided doses (35 mg elemental iron/300 mg ferrous gluconate)	GI: nausea, dyspepsia, constipation and/or diarrhea. Ameliorated by stepwise initiation of therapy (see text).	There is no evidence that one preparation is more effective than another. Avoid enteric-coated preparations. Vitamin C may enhance absorption of nonheme iron. Absorption is decreased by food and certain drugs: antacids, calcium

Drug/Cost[a]	Dosage	Adverse Effects	Comments
			carbonate, cholestyramine, levodopa, methyldopa, penicillamine, quinolones, sodium bicarbonate, tetracyclines. Separate administration by ~2 h.
ferrous sulfate Fer-In-Sol, generics < $5	105–200 mg elemental iron daily in 3 divided doses (60 mg elemental iron/300 mg ferrous sulfate)	GI: nausea, dyspepsia, constipation and/or diarrhea. Ameliorated by stepwise initiation of therapy (see text).	There is no evidence that one preparation is more effective than another. Avoid enteric-coated preparations. Vitamin C may enhance absorption of nonheme iron. Absorption is decreased by food and certain drugs: antacids, calcium carbonate, cholestyramine, levodopa, methyldopa, penicillamine, quinolones, sodium bicarbonate, tetracyclines. Separate administration by ~2 h.
polysaccharide-iron complex FeraMAX, Polyride-Fe, Triferexx, generics $10–15	105–200 mg elemental iron daily (150 mg elemental iron per capsule)	GI: nausea, dyspepsia, constipation and/or diarrhea. Ameliorated by stepwise initiation of therapy (see text).	There is no evidence that one preparation is more effective than another. Avoid enteric-coated preparations. Vitamin C may enhance absorption of nonheme iron. Absorption is decreased by food and certain drugs: antacids, calcium carbonate, cholestyramine, levodopa, methyldopa, penicillamine, quinolones, sodium bicarbonate,

Drug/Cost[a]	Dosage	Adverse Effects	Comments
			tetracyclines. Separate administration by ~2 h.
Drug Class: Iron Supplements, parenteral			
ferric carboxymaltose Ferinject $50/100 mg vial	Total IV dose calculated based on body weight and hemoglobin Can be administered as an IV infusion or as an IV bolus injection	Anaphylaxis (rare). Infusion reactions, hypotension. Headache, edema, injection site reactions.	Consult prescribing information for dose calculation, dilution and administration details. 0.9% NaCl is the preferred infusion solution.
ferric derisomaltose Monoferric $50/100 mg vial	Total IV dose calculated to restore iron deficit in red cell mass and iron stores Can be administered as an IV infusion or as an IV bolus injection Chronic dialysis patients: 100 mg 1–3 times weekly IV × 10 doses or until replete; adjust as able to maintain adequate iron availability	Anaphylaxis (rare). Infusion reactions, hypotension. Fever, chills, headache, myalgia, arthralgia, urticaria, dizziness.	Consult prescribing information for dose calculation, dilution and administration details. 0.9% NaCl is the preferred infusion solution. Side effects are more common with larger doses and in underweight patients.
iron sucrose Venofer, generics $60/100 mg vial	Chronic dialysis patients: 100 mg 1–3 times weekly IV until replete; adjust as able to maintain adequate iron availability Administered as IV infusion or slow injection	Anaphylaxis (rare). Infusion reactions, hypotension (avoid rapid IV infusion). Fever, chills, headache, myalgia, arthralgia, urticaria, dizziness.	Test doses are not recommended by the manufacturer. Incidence of life-threatening adverse effects is lower than with iron dextran.

Drug/Cost[a]	Dosage	Adverse Effects	Comments
sodium ferric gluconate Ferrlecit $60/62.5 mg vial	Chronic dialysis patients: 125 mg 1–3 times weekly IV × 8 doses; adjust as able to maintain adequate iron availability Administered as IV infusion over 1 h or slow IV injection (maximum rate of 12.5 mg/min)	Anaphylaxis (rare). Infusion reactions, hypotension (avoid rapid IV infusion). Fever, chills, headache, myalgia, arthralgia, urticaria, dizziness. May have delayed onset.	Test doses are no longer recommended by the manufacturer. Incidence of life-threatening adverse effects is lower than with iron dextran.
Drug Class: Vitamins			
folic acid generics < $5	0.4–5 mg daily PO	Occasional allergic reactions: rash, pruritus, flushing, bronchospasm.	Available for IV use in patients who are fasting. Empiric supplementation of folate in patients who are at risk of B_{12} deficiency and not receiving B_{12} is not recommended. For pregnancy 0.4 mg/day beginning at least 2–3 months before conception and continuing until 4–6 wk postpartum or as long as breastfeeding continues is recommended. Higher doses recommended when risk of neural tube defect is higher (see Table 1, Nutritional Supplements).
(cyanocobalamin, hydroxocobalamin) vitamin B_{12} generics $5–10	Pernicious anemia/other chronic malabsorption disorders: 100 mcg daily SC/IM × 1 wk; 200 mcg weekly SC/IM until Hb normalizes Lifelong maintenance: 100 mcg monthly SC/IM *or* 1000–2000 mcg daily PO Vitamin B_{12} deficiency: 30 mcg daily SC/IM × 5–10 days *or* 500–2000 mcg daily PO Lifelong maintenance: 100–200 mcg monthly SC/IM *or* 250 mcg daily PO	Occasional peripheral vascular thrombosis, rash, pruritus, headache, nausea and vomiting, diarrhea. Occasional heart failure. Possibility of profound hypokalemia due to intracellular potassium shift.	Oral absorption is reduced by anticonvulsants, colchicine, metformin, neomycin and PPIs. Consider baseline measurement and potassium supplementation if levels are low or borderline; serial monitoring of potassium levels may be warranted. Folate supplementation may mask hematologic findings of B_{12} deficiency without halting progression of the neurologic deficits. Vitamin B_{12} should be given parenterally (IM or SC) in patients with documented malabsorption or in patients who are nonadherent to oral therapy.

[a] Cost of 30-day supply unless otherwise specified; includes drug cost only.

Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.

BMI = body mass index; GI = gastrointestinal; PPI = proton pump inhibitor

Table 3: Erythropoiesis-Stimulating Agents

Drug/Cost[a]	Dosage	Adverse Effects	Comments
Drug Class: Erythropoietics			
darbepoetin alfa Aranesp $285/100 mcg	Chronic renal failure: 0.45 mcg/kg weekly SC/IV, then increase by 25% monthly if no response; decrease by 25% as Hb approaches 120 g/L Cancer chemotherapy (endogenous erythropoietin level ≤200 units/L): 2.25 mcg/kg weekly SC/IV	Hypertension, hypotension, headache, thrombosis, nausea, vomiting, diarrhea, constipation, arthralgia, myalgia, chest pain, arrhythmia, edema, dyspnea, cough.	May be able to shift to biweekly or monthly dosing in some patients. Target Hb ≤120 g/L with increase limited to 10 g/L/2 wk. If excessive response, decrease dose by 40%. If still excessive, hold dose until Hb falls.

Drug/Cost[a]	Dosage	Adverse Effects	Comments
	If inadequate response after 6 wk, increase to 4.5 mcg/kg weekly SC/IV		
epoetin alfa Eprex $150/10 000 units	Chronic renal failure: Initial: 50–100 units/kg 3 times weekly SC/IV, then increase by 25% every 4–8 wk to maximum 300 units/kg/dose to achieve Hb of 120 g/L As Hb approaches 120 g/L, decrease dose by 25% HIV, on antiretrovirals (endogenous erythropoietin level ≤500 units/L): Initial: 100 units/kg 3 times weekly SC/IV, then increase by 50 units/kg/dose every 4–8 wk to maximum 300 units/kg/dose Cancer chemotherapy (endogenous erythropoietin level ≤200 units/L): Initial: 50 units/kg 3 times weekly SC/IV, or 40 000 units weekly SC/IV, then increase by 50 units/kg/dose every 8 wk to maximum 300 units/kg/dose Chronic hepatitis C, on ribavirin: 40 000 units weekly SC/IV Surgery: 600 units/kg SC/IV 21, 14 and 7 days before surgery and then on the day of surgery	Hypertension, headache, seizures, thrombosis, nausea, vomiting, diarrhea, arthralgia, chest pain, edema, cough. Increases risk of deep venous thrombosis and other thrombotic complications in spine surgery patients. Pure red cell aplasia (rare).	Usually given by SC injection. May be given IV if access already established. Do not exceed target Hb. If no response to the maximum dose after 8 wk, discontinue. If Hb increases by more than 10 g/L per 2 wk, decrease dose by 25%. Target Hb ≤120 g/L. In patients with renal failure, achievement of higher Hb targets is not associated with better outcomes.[62][63][64] Survival was worse in patients treated to higher Hb targets (>120 g/L) with erythropoietin as compared with placebo in patients with head and neck cancer and in those with breast cancer.[65][66]

[a] Cost per dose; includes drug cost only.

Suggested Readings

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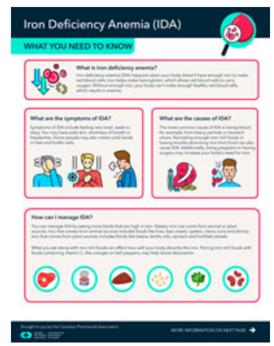
Information for the Patient

- Iron Deficiency Anemia (IDA)

Infographic – Iron Deficiency Anemia (IDA)



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